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Project Title: Transjakarta: Jakarta Public Bus Interconnectivity and Coverage

Introduction

In March 2024, Seasia Stats ranked Transjakarta as the world's longest Bus Rapid Transit (BRT) system (Seasia, 2024). Despite this achievement, Greater Jakarta's growing urban population is projected to become the largest urban agglomeration by 2030 (Jauregui-Fung, 2022) –poses challenges in maintaining efficient public transport interconnectivity and accessibility. To ensure Transjakarta remains effective amid increasing demand, data-driven insights are crucial for guiding future improvements.

Motivation

As a regular Transjakarta user, I have personally experienced its reliability in improving urban mobility. However, I want to explore how well Transjakarta accommodates public demand and **identify areas that need optimization and expansion**. By analyzing its **route efficiency, user profiles, and infrastructure readiness**, I aim to **provide insights that can support policy-making and urban transit planning for Jakarta** and other cities developing public transport networks.

Questions

1. Do current bus stops and routes effectively accommodate user demand?
2. What are the user profiles for each route, and how can this inform future service improvements?
3. Which bus stops require upgrades or renovations to handle growing passenger numbers?

Data sources

- A. Transjakarta - Public Transportation Transaction**, April 2023. Tabular data in CSV format with 189.500 rows x 22 columns. Contains transaction details, including transaction ID, user profile (ID, name, payment details, gender, birth year), and travel details (tap-in and tap-out bus stops, routes, latitude and longitude, timestamps, and payment amount).
(<https://www.kaggle.com/datasets/dikisahkan/transjakarta-transportation-transaction/>)
- B. Transjakarta Greater Jakarta Location Data**, updated Jan 2023. Tabular data in HTML format with 210 rows x 7 columns. Contains bus stop locations, corridor names and routes, year built, data period, and remarks.
(https://satudata.jakarta.go.id/open-data/detail?kategori=dataset&page_url=data-lokasi-transjakarta-dki-jakarta&data_no=1)

Both datasets will be used together to complement each other and provide a more comprehensive analysis to answer all research questions.

References

- Jauregui-Fung, F. (2022). *BRT Transjakarta: Phasing in, performing and expanding a new system within a consolidated urban area: Report for the “Inclusive and sustainable smart cities in the framework of the 2030 Agenda for Sustainable Development” Project*. German Institute of Development and Sustainability (IDOS).
<https://doi.org/10.23661/r6.2022>
- Seasia Stats. (2024, March). *The world's longest bus rapid transit (BRT) systems* [Instagram post]. Instagram.
<https://www.instagram.com/seasia.stats/p/C6h4kBRSCDy/>